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ASSIGNMENT-

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Subject: DBMS

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Ans-1) Database security issues refer to the problems & risks related to protecting a database from unauthorized access, misuse or data leaking. These issues are mainly categorized into three types-

1) Legal & Ethical issues

Legal & ethical issues concern the proper use and protection of data according to laws and moral responsibilities.

Legal issues → Legal issues arise from laws and regulations related to data protection. Organizations must follow these laws when collecting, storing or sharing data.

eg → → Personal data must not be shared without permission
→ companies must protect customer data

2) Policy issues

Policy issues refer to organizational rules and guidelines created to ensure database security.

These policies define

- Who can access the database
- Who can modify data
- Who can delete data.

Examples of security policies

- Password policies
- User authetication policies
- Data backup policies
- Access level policies

Such policies help prevent unauthorized access to the database

3) System-Related Issues

System-related issues arise due to technical weakness or vulnerabilities in the database system

Examples → Weak password
Software bugs
Hacker attacks
Malware or viruses

To prevent system-related issues, techniques such as
- Firewalls
- Authetication mechanisms
- Secure configuration
- Regular system update

Ans-2) Database security control techniques are methods used to protect the database from unauthorized access and misuse.

1) Access Control

Access control determines which users are allowed to access specific data and what operations they can perform. It defines permissions such as → Read, Insert, Update

Example →

User	Permission
Admin	Full access
Employee	Read only
Manager	Read & update

SQL example

```
GRANT SELECT ON Employee To user1;
```

Inference Control

Inference control prevents users from deducing sensitive information indirectly from available data

eg→ If someone knows

- The average salary in department
- Only one employee works in that department

Then the person's salary can be easily guessed

~~Inf~~ Inference control methods prevent such situation by

- Restricting queries
- Limiting access to detailed data

3) Flow Control

Flow control ensures that information does not flow from a higher security level to a lower security level in an authorized way

eg→ • Top secret • Secret • Public

This helps prevent data leakage between different security level.

4) Data Encryption →

Data Encryption is the process of converting readable data into an encoded form so that unauthorized users cannot understand it

eg→ Original data

salary = 50000

Encrypted data

X7A9#L2P

Only users with the correct decryption key can convert the data back to its original form

Encryption is commonly used in

- Online Banking
- Password storage
- Secure database systems

Ans- 3 SQL View Creation

Given table

Employee (EmpID, EmpName, salary, DeptID)

We need to create a view that displays only EmpID, EmpName and DeptID

SQL Query

```
CREATE VIEW Employee_view AS
SELECT EmpID, EmpName, DeptID
FROM Employee;
```

Explanation →

→ CREATE VIEW → creates a virtual table

Employee_View → name of the view

SELECT EmpID, EmpName, DeptID → selects specific columns

FROM Employee → data comes from the Employee table

To view the data

```
SELECT * FROM Employee_View;
```

This view will hide the salary column and display only the selected fields